



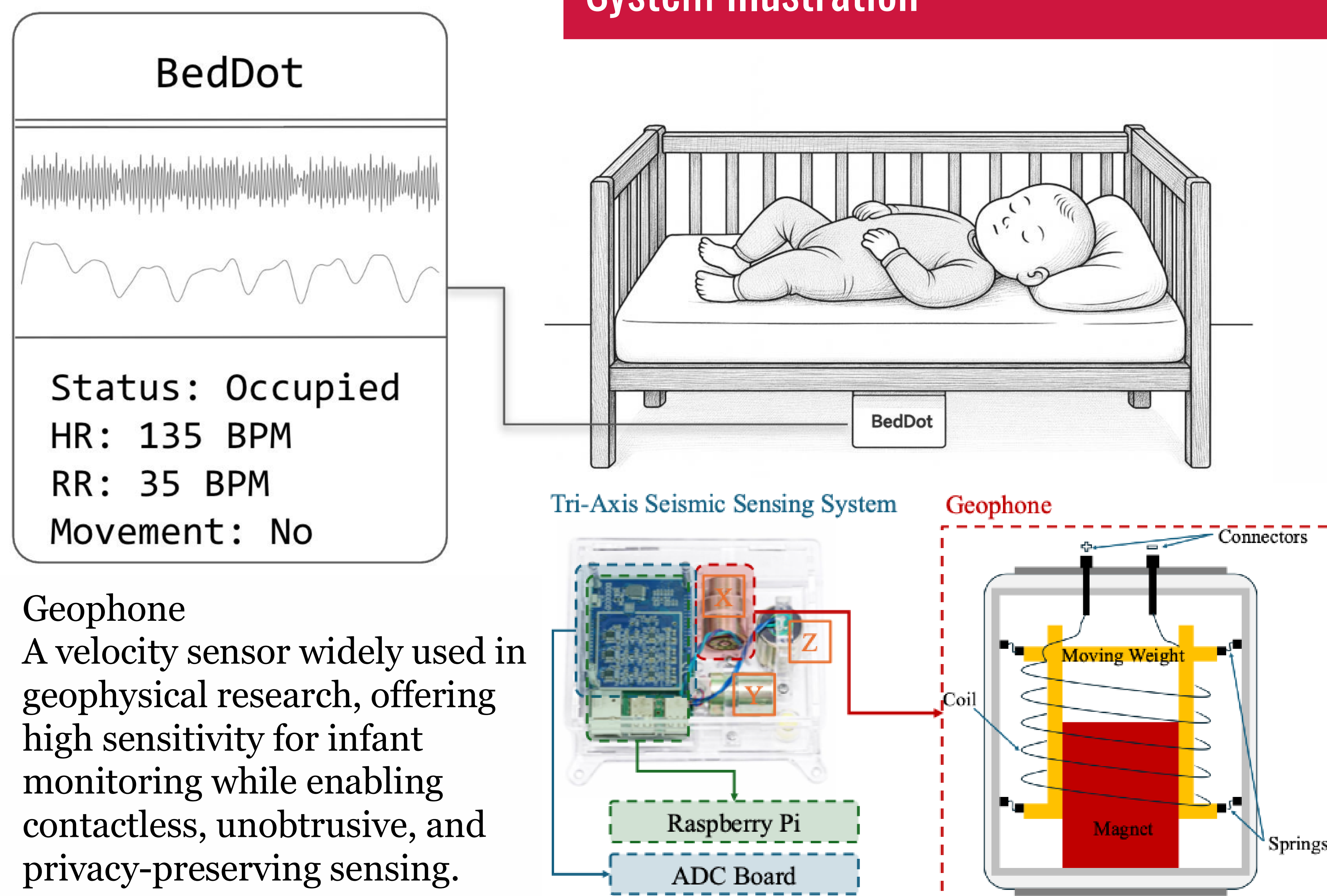
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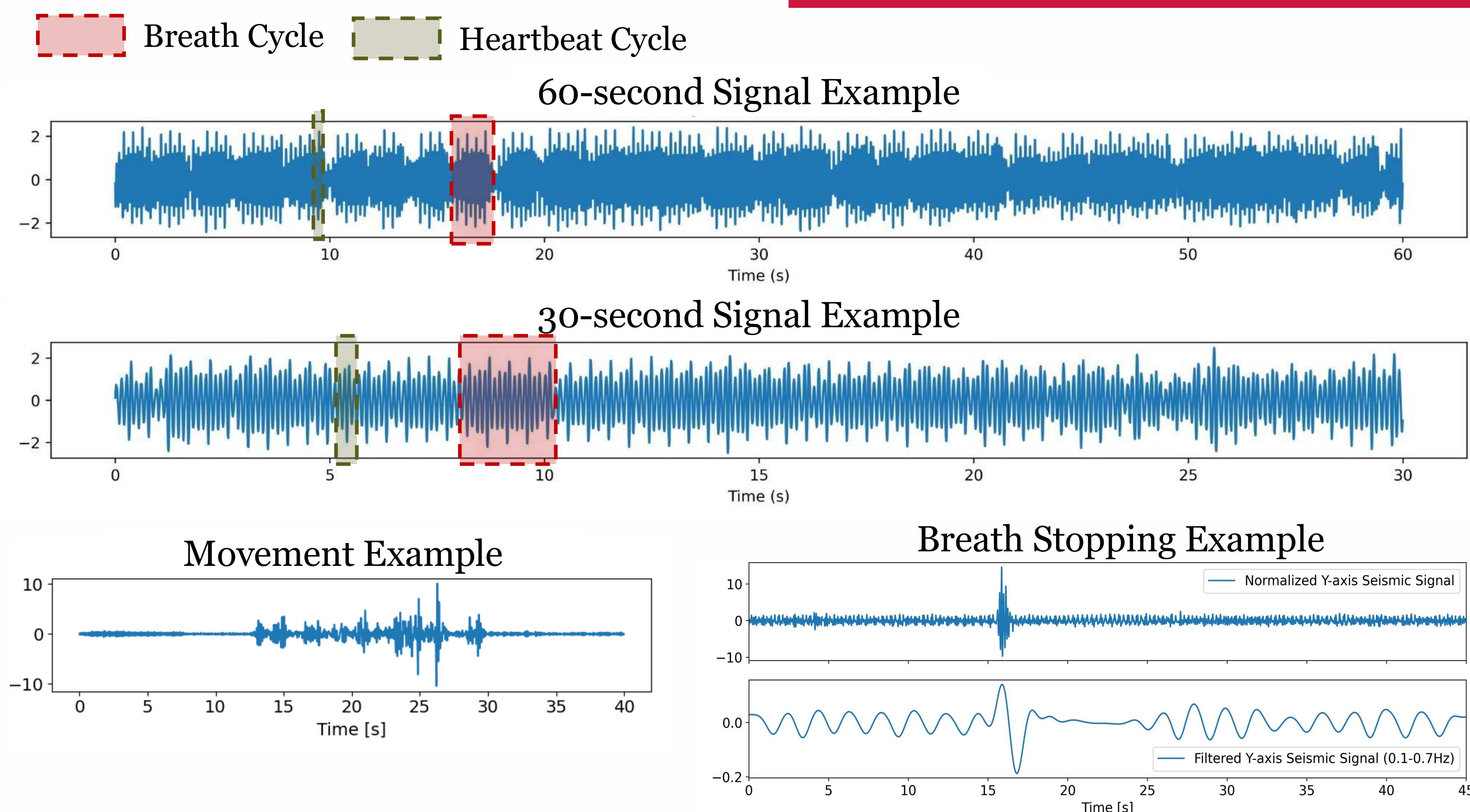
Non-Contact, Unobtrusive, and Privacy-Preserving Infant Vital Signs Monitoring

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Sensorweb, School of Electrical and Computer Engineering, University of Georgia

System Illustration



Signal Characterization

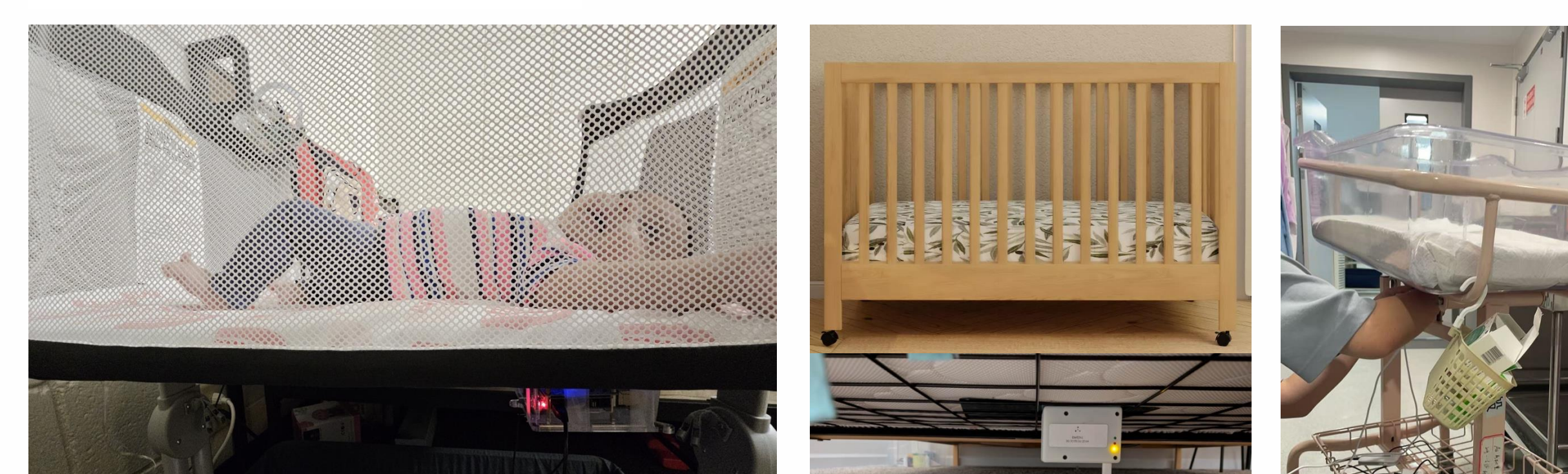


Whole Night Monitoring



Preliminary Validation

Data Collection

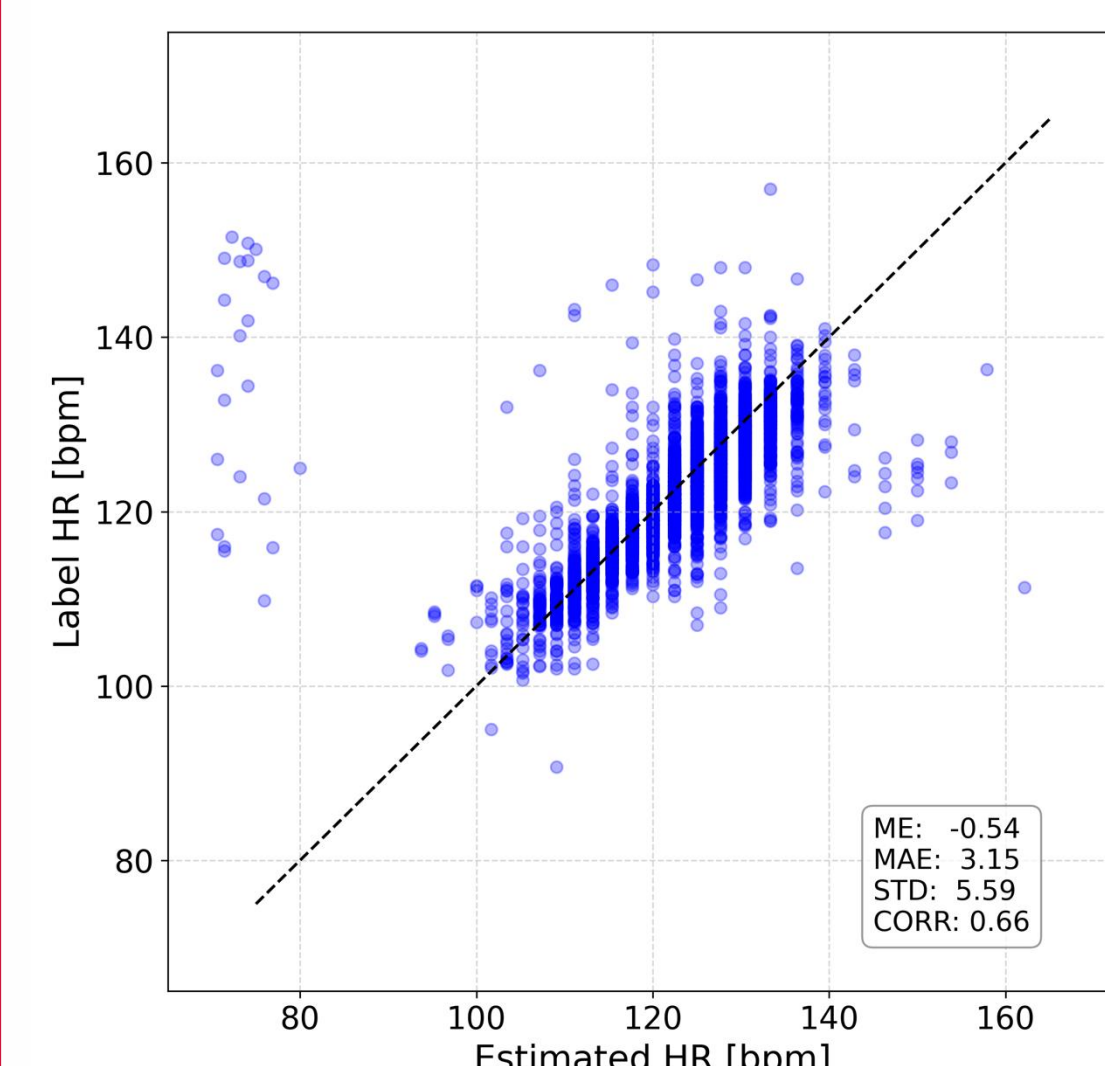


Bassinet

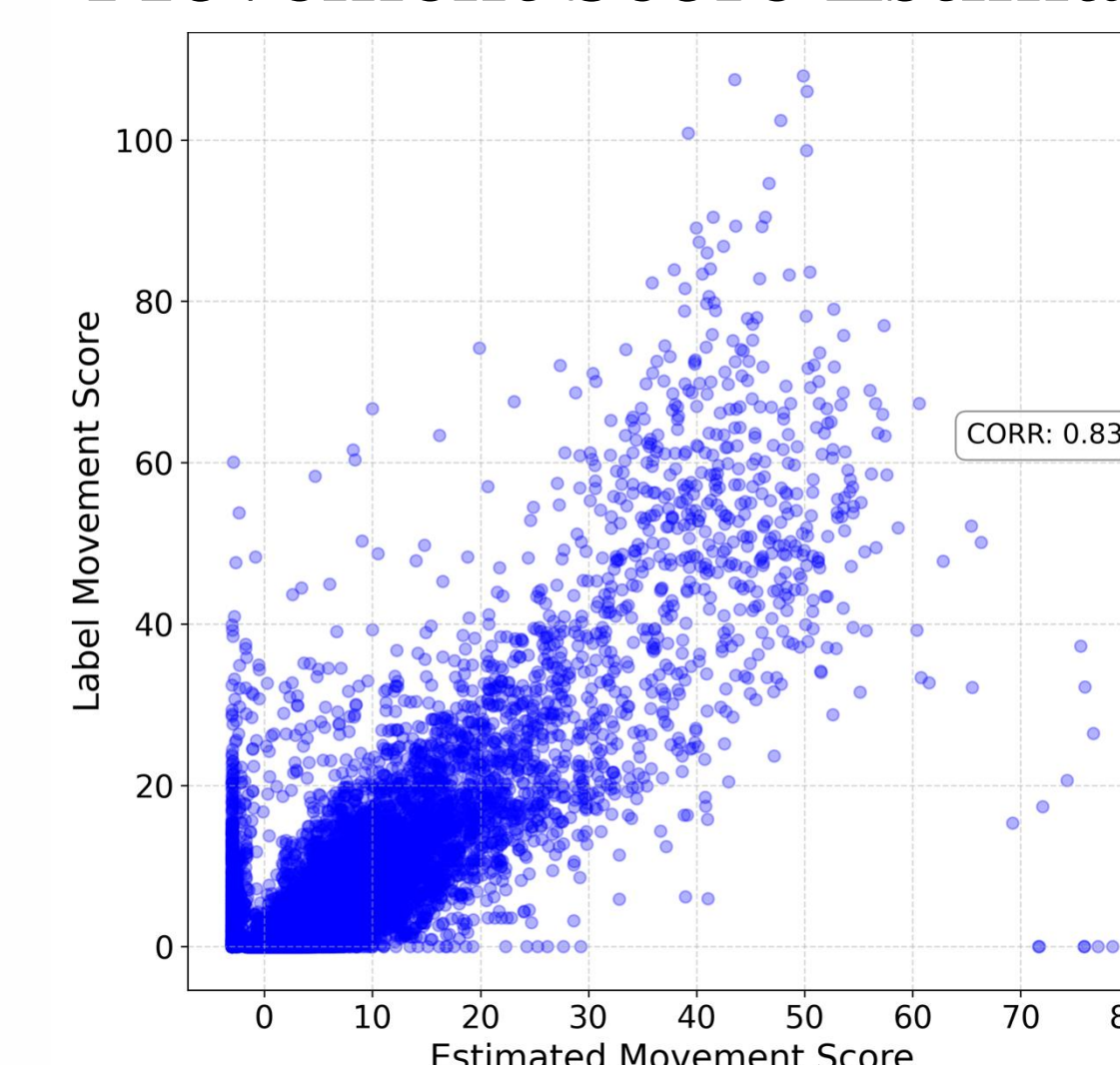
Incubator

Crib

Heart Rate Estimation



Movement Score Estimation



Future Works



Sleep Stage
Classification

Breath Stop
Detection

Crying
Detection

Key Takeaways

Contactless sensing using geophones
Robust occupancy detection
Accurate HR & RR estimation
Suitable for long-term infant monitoring